E-DIAGNOSTIC LAB MANAGEMENT SYSTEM

PROJECT REPORT

Submitted by

NAME OF STUDENT

[EA23322*****6.]**

Under the Guidance of

Dr.G.Babu

(Assistant Professor, Directorate of Online Education)

In partial fulfillment for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

DIRECTORATE OF ONLINE EDUCATION
SRM INSTITUTE OF SCIENCE & TECHNOLOGY

KATTANKULATHUR- 603 203

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DIRECTORATE OF ONLINE EDUCATION

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

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BONAFIDE CERTIFICATE

This Project report titled **"E-Diagnostic Lab Management System"** is the Bonafide work of **Name of student [EA233*********6]** who completed the Project work along with the company mentor. Certified further, that to the best of my knowledge the work reported herein does not form any other Project report or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidate

Signature of Student

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PROJECT OFFER LETTER

This is to certify that **NAME OF STUDENT [EA233**********6]** has undergone Project work during the period from 04-02-2025 to 28-04-2025 under the guidance of S. Sureshkumar.

This work has been carried out in partial fulfillment of the requirements for the award of the degree of **MASTER OF COMPUTER APPLICATIONS**

For Brillio Technologies Pvt. Ltd.

Jayarajan C

BRILLIO TECHNOLOGIES PVT. LTD. BANGALORE

Jayarajan C
Manager – Human Resources

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ABSTRACT

ABSTRACT

It is an online diagnostic lab manager application that brings up various diagnoses working online. Here patients get an option to register on the site and login using registered details. Once registered with their address and contact details, the patients may now see a variety of tests conducted by the lab along with their costs. The system allows patients to do the bookings for various tests like CBC, Blood Glucose, KFT, and LFT. The tests also consist of parameters like Hemoglobin, WBC, etc.

The system allows users to book any test needed and after successful booking, system calculates costs and allows users to pay online. As soon as the payment is done, the patient test is booked and the lab may now collected samples from patients registered address. After successful testing, the patient or the user gets a notification of test result through an email.

The system allows admin to attach The system allows patients to do the bookings for various tests like CBC, Blood Glucose, KFT, and LFT. The tests also consist of parameters like Hemoglobin, WBC, etc. a copy of the report into the system and automatically email it to intended patient.

INTRODUCTION

1. INTRODUCTION

The program is an on the Diagnostic Lab Reporting System administrator program which provides various analysis working on the internet. Here victims are first able to sign-up on the site and also sign in using authorized information. Once authorized with their deal with and contact information, the patient may now see a wide range of assessments performed by the lab along with their expenses. The program allows for CBC, Blood Sugar, KFT, LFT assessments to be reserved by individual. The assessments also comprises of factors like Hemoglobin, WBC, etc. Now the program allows customers to book any analyze needed. After effective reservation program determines expenses and allows customers to pay online.

1.1 Objectives

Technology has facilitates human beings in almost every field of life. They turn manual tasks automatic to saves recourses. Automatic works is considered more trustful, reliable, accurate etc. Technology is the main reason which successfully manages many processes and creates successful management system.

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PROBLEM DESCRIPTION

2. PROBLEM DESCRIPTION

The online diagnostic lab reporting system was developed for all diagnostic centres to replace the existing manual paper-based system. In the new system, they can control the following information: Patient information, staff information, and diagnostic laboratory information system. These services need to be delivered in an efficient and cost-effective manner with the goal of reducing the time and resources currently required for such tasks. What is an online diagnostic test reporting system? The important thing in operating a diagnostic centre is to collect, manage, and acquire a large amount of information in a timely manner. This information is typically related to patient personal information, staff information, lab information including time and results, and payment history.

1 .1 Module Description

System Users and Their Functions:

3.2 System Features

In the life of the software development, problem analysis provides a base for design and development phase. The problem is analyzed so that sufficient matter is provided to design a new system. Large problems are sub-divided into smaller once to make them understandable and easy for finding solutions. Same in this project all the task are sub-divided and categorized.

System Modules:

- **Admin**
- **Patients**

Module Description:

Registration:

The patients or clients who are using the proposed website have to register them on the website by providing basic details about them, their medical details and contact details. They can set their unique name and password.

Login:

The users (patients or clients) access the system or system features by login into the website by using their username and password.

ViewInterface:

On main or home page of the proposed website the section of tests conducting is reserved. On that page each test along with their estimated cost is displayed for example CBC, Blood Glucose, KFT etc.

LocateNearbyLab:

The system locates the nearby lab and their working hours as well the time slots available to them.

BookTestorReserving:

The client can book certain needed test online. Client can take available slots or in case reserve demanded slots up to requirements.

Cost Calculation and Payment:

After successful booking the system calculates the estimated cost and the user have to pay online.

LabReceive:

The person's tests are conducted in nearby lab or any unit mentioned by lab and collected from their later.

TestsResults:

After the test is conducted and samples are examined the patient gets reports results of their test with the help of notification. The users can immediately view their reports and can collect hard copy of reports from diagnostic center later.

Tools/Platform
Hardware and Software
Requirement specifications.

3 Tools/Platform- Hardware and Software Requirement specifications

3.1 Android Tools:

IDE: Android Studio

Android Emulator

Tools Requirements:-Android studio Android emulator

PHP tools :-XAMPP

3.2 Hardware Requirements:-

Processor	:	Intel(R) 2.10GHz
Installed memory (RAM)	:	4 GB
Hard Disk	:	160 GB
Operating System	:	Windows (7)

3.3 Software Requirements:-

Front End: HTML5, CSS3, Bootstrap Back

End: PHP 7.4, MYSQL

Control End: Angular Java Script

PROBLEM DEFINTION

4 Function Requirements– DESIGN

Design is the first step in the development phase for any techniques and principles for the purpose of defining a device, a process or system in sufficient detail to permit its physical realization.

Once the software requirements have been analyzed and specified the software design involves three technical activities - design, coding, implementation and testing that are required to build and verify the software.

The design activities are of main importance in this phase, because in this activity, decisions ultimately affecting the success of the software implementation and its ease of maintenance are made. These decisions have the final bearing upon reliability and maintainability of the system. Design is the only way to accurately translate the customer's requirements into finished software or a system.

Design is the place where quality is fostered in development. Software design is a process through which requirements are translated into a representation of software. Software design is conducted in two steps. Preliminary design is concerned with the transformation of requirements into data.

a. UML Diagrams:

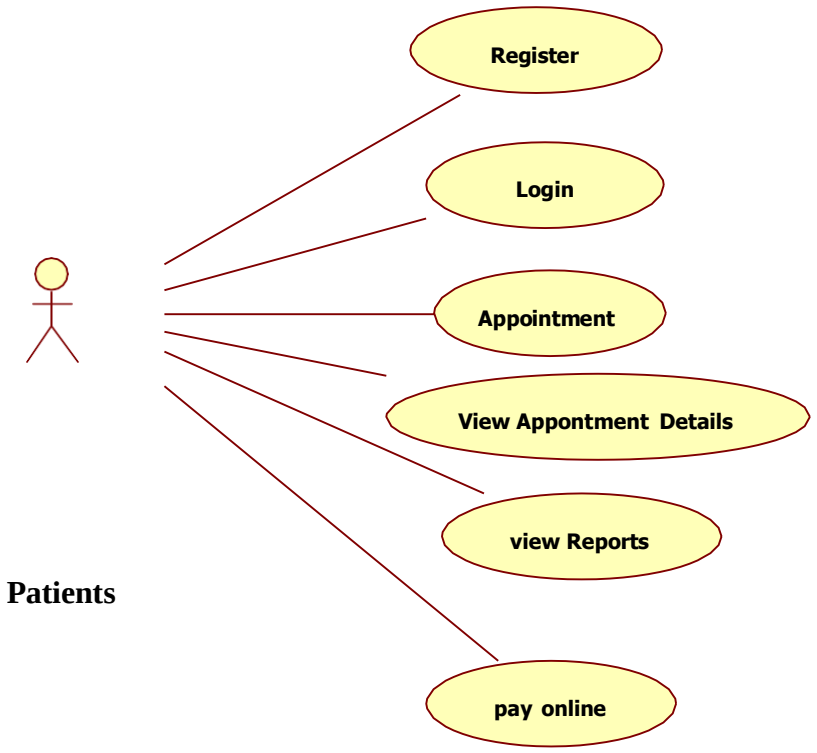
UML stands for Unified Modeling Language. UML is a language for specifying, visualizing and documenting the system. This is the step while developing any product after analysis. The goal from this is to produce a model of the entities involved in the project which later need to be built. The representation of the entities that are to be used in the product being developed need to be designed.

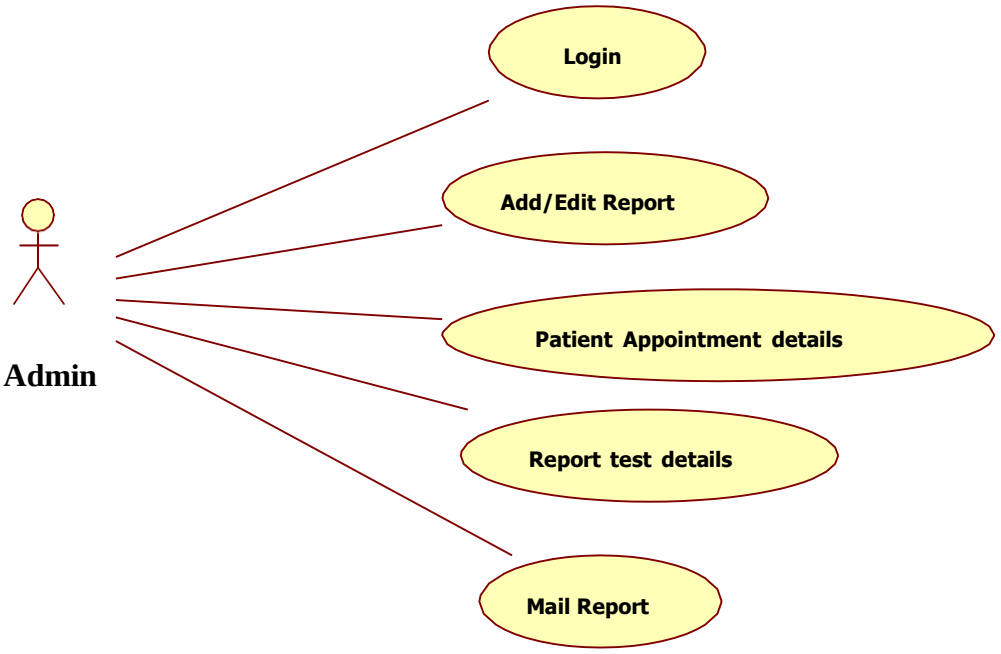
There are various kinds of methods in software design:

- Use case Diagram
- Sequence Diagram
- Collaboration Diagram

i. Usecase Diagrams:

Use case diagrams model behavior within a system and helps the developers understand of what the user require. The stick man represents what's called an actor. Use case diagram can be useful for getting an overall view of the system and clarifying who can do and more importantly what they can't do.





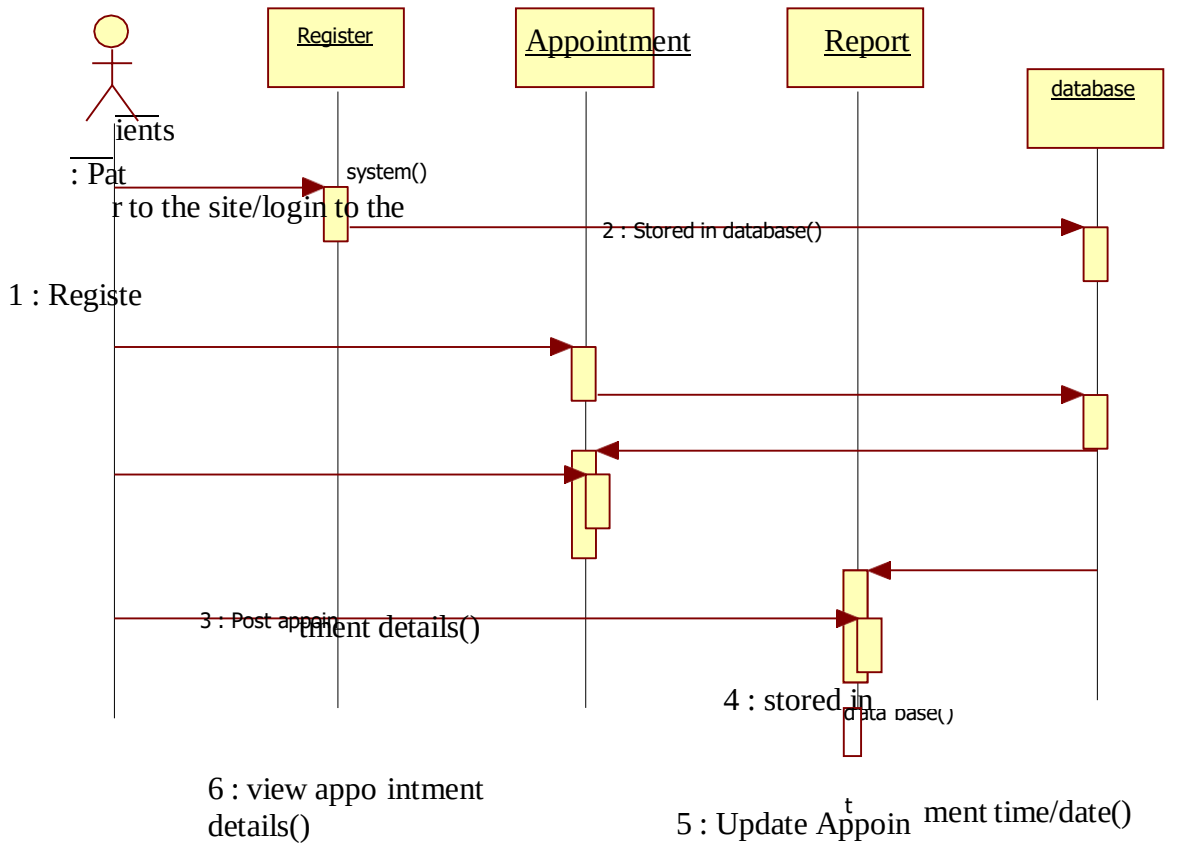
Use case diagram consists of use cases and actors and shows the interaction between the use case and actors.

1. The purpose is to show the interactions between the use case and actor.
2. To represent the system requirements from user's perspective.
3. An actor could be the end-user of the system or an external system

ii. Sequence Diagram:

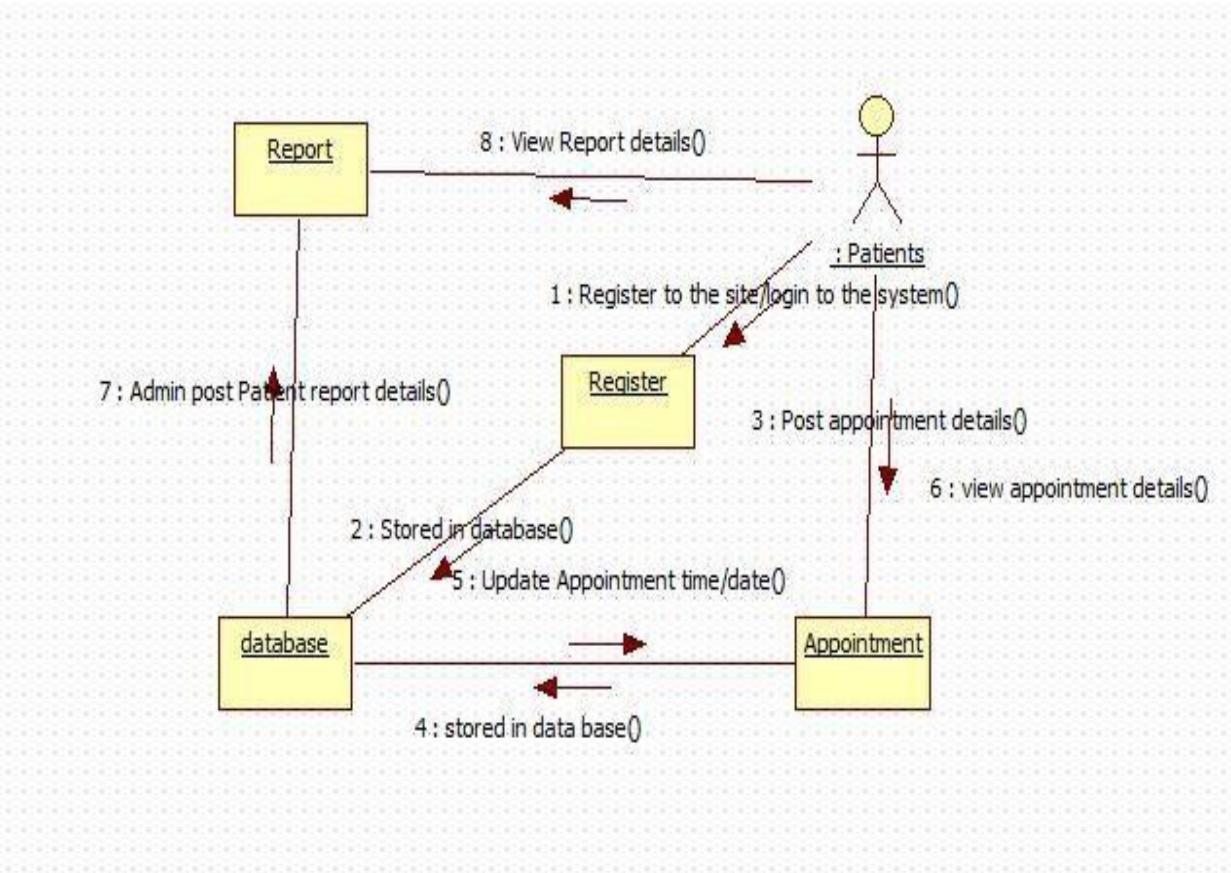
Sequence diagram and collaboration diagram are called INTERACTION DIAGRAMS. An interaction diagram shows an interaction, consisting of set of objects and their relationship including the messages that may be dispatched among them.

A sequence diagram is an introduction that empathizes the time ordering of messages. Graphically a sequence diagram is a table that shows objects arranged along the X-axis and messages ordered in increasing time along the Y- axis.



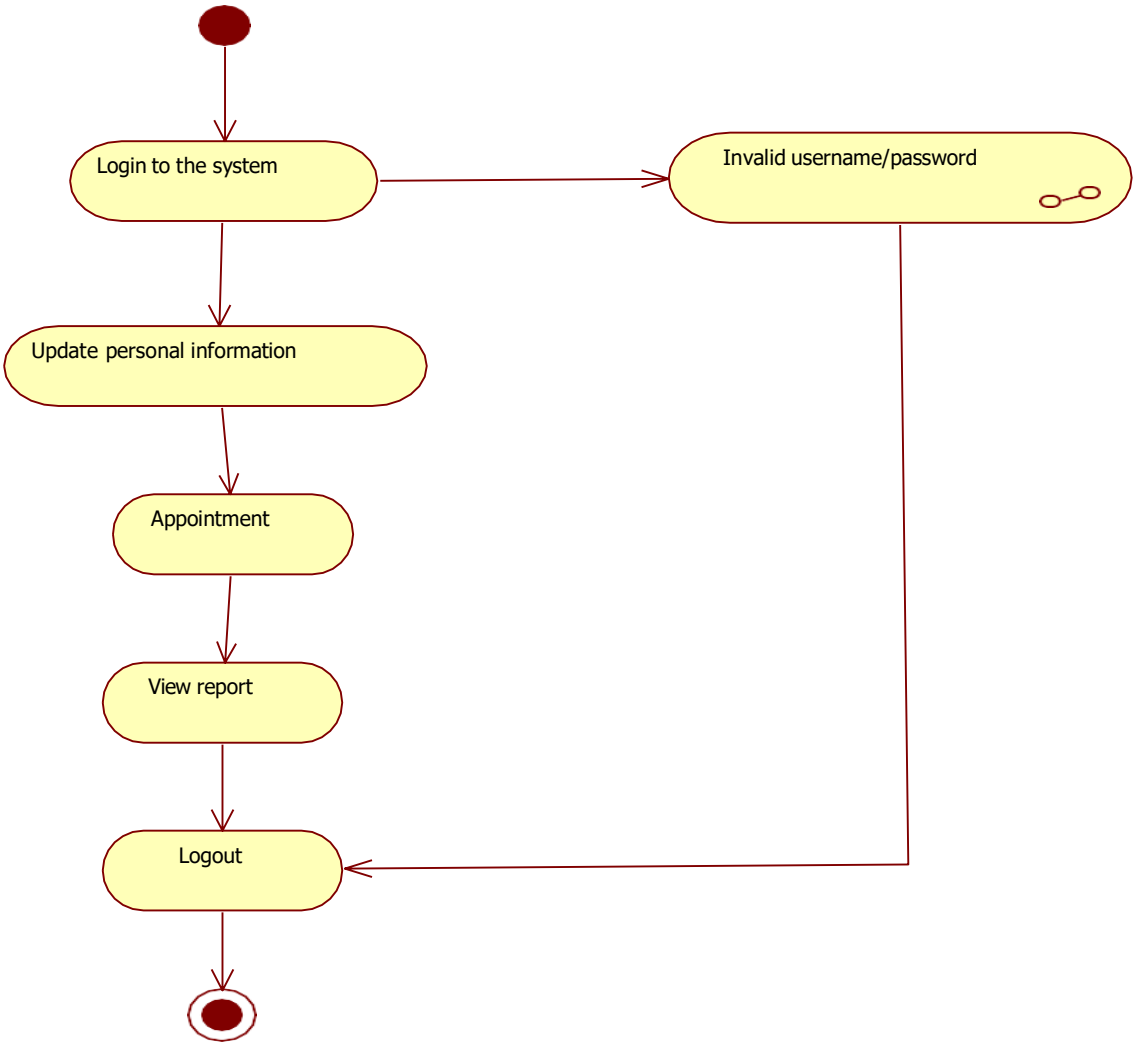
iii. Collaboration Diagram:

A **collaboration diagram** is a type of visual presentation that shows how various software objects interact with each other within an overall IT architecture and how users can benefit from this **collaboration**. A **collaboration diagram** often comes in the form of a visual chart that resembles a flow chart.

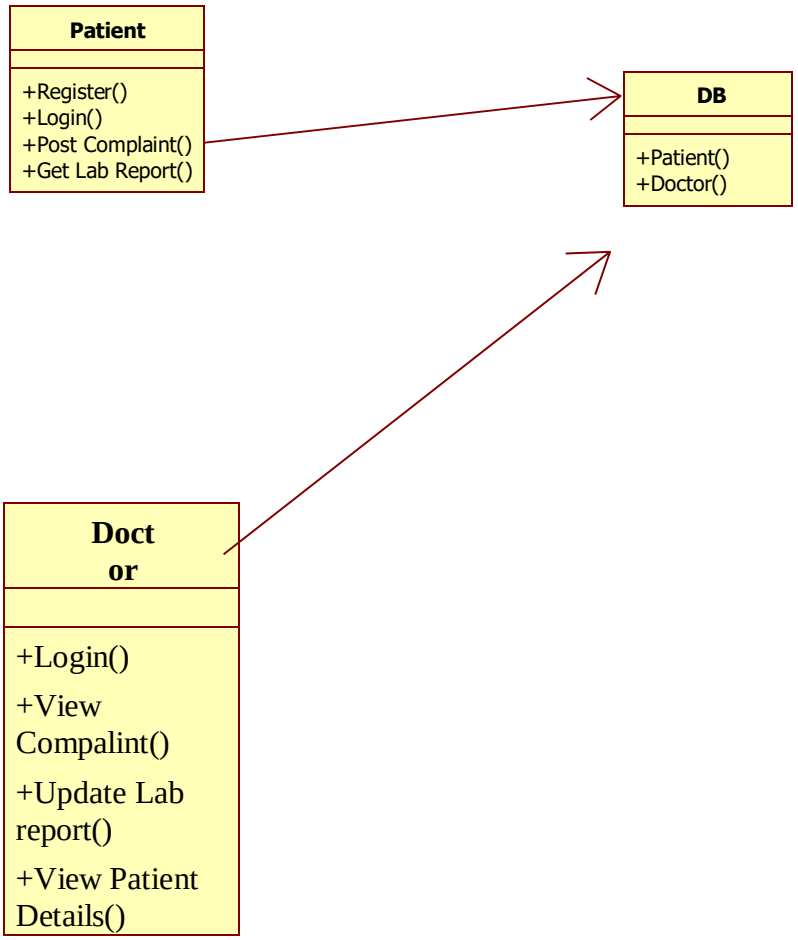


iv. Activity Diagram

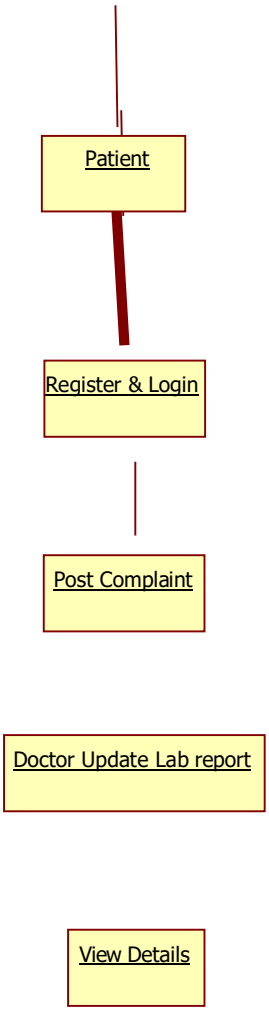
Activity diagram is another important diagram in UML to describe dynamic aspects of the system. Activity diagram is basically a flow chart to represent the flow from one activity to another activity. The activity can be described as an operation of the system. So the control flow is drawn from one operation to another.



v. Class Diagram



vi. DFD Diagram



vii. Table Design

User Register & Login

User ID	Name	Email Id	Password	Mobile

Doctor Table

User ID	Name	Email Id	Password	Mobile	Department

Lab Data Table

User ID	Name	Father Name	Age	DOB	Sympton	Problem	Doctor	Department

Lab Soltuion Table

User ID	CBD	Blood Glucose	KFT/LFT	Hemoglobin	WBC

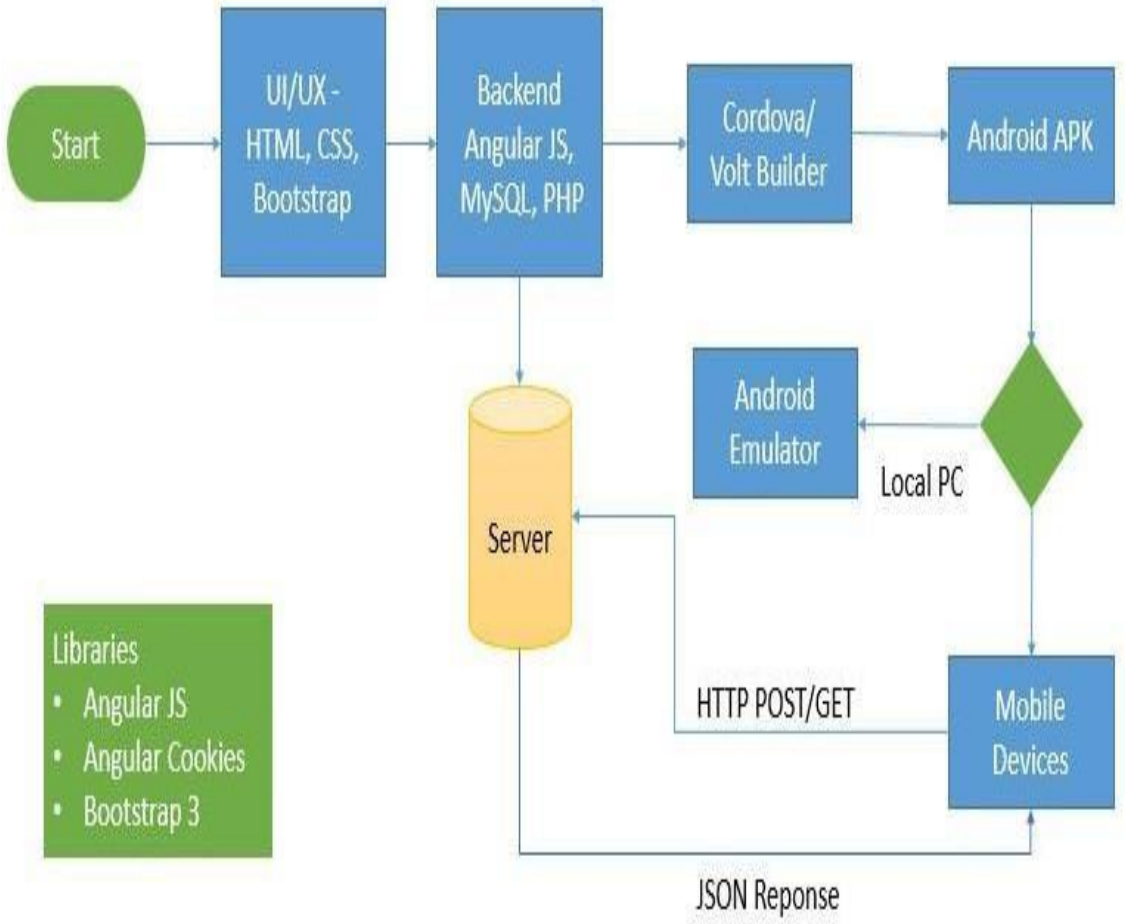
Cashier

User ID	Eamil Id	Password	

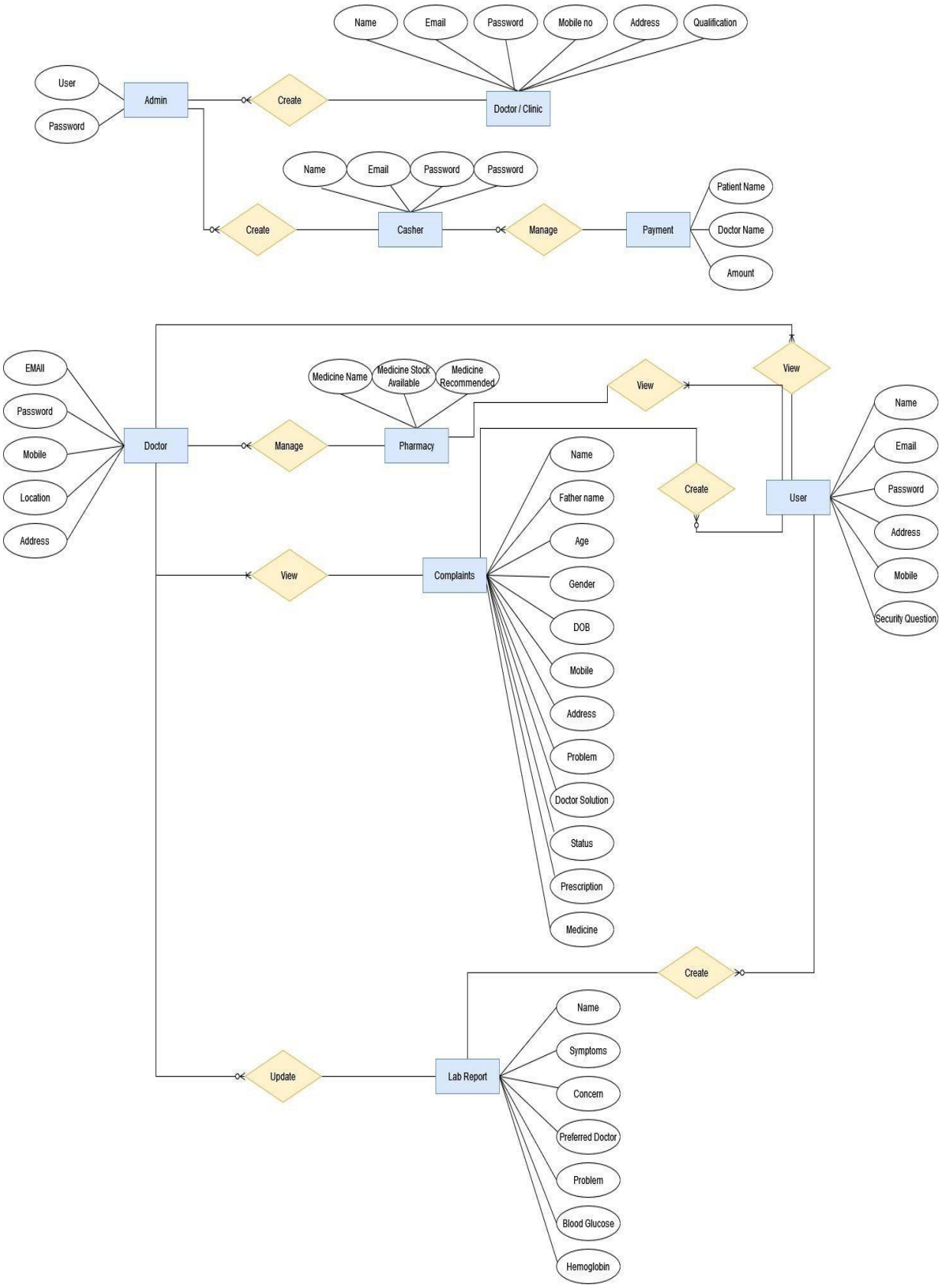
Payment Table

User ID	Patient Name	Doctor	Amount

viii. WORKFLOW DIAGRAM



ER DIAGRAM


Name
Email
Password
Mobile no
Address
Qualification
User
Password
Admin
Create
Doctor / Clinic
Name
Email
Password
Password
Patient Name
Create
Casher
Manage
Payment
Doctor Name
Amount
EMAll
Password
Mobile
Location
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Medicine Name
Medicine Stock Available
Medicine Recommended
View
View
Name
Email
Password
Address
Mobile
Security Question
Doctor
Manage
Pharmacy
Name
Father name
Age
Gender
DOB
Mobile
Address
Problem
Doctor Solution
Status
Prescription
Medicine
Create
User
View
Complaints
Create
Name
Symptoms
Concern
Preferred Doctor
Problem
Blood Glucose
Hemoglobin
Update
Lab Report

SCOPE OF ANALYSIS AND SOLUTION

4 SCOPE OF ANALYSIS AND SOLUTION

Existing Solution:

It needs employment as the human efforts are being automated by this system.

In this existing Diagnostic Lab Reporting System, the reports are generated either manually or by using computer software in which the details itself needs to be fitted by the employees of the organization. Existing system has greater tendency of having more errors as compared to the automation system. In existing Diagnostic Lab Reporting System, as the chances of errors are greater, thus the chances of getting information from the reports are more which will indirectly or directly affect the patients and their health only.

Existing Diagnostic Lab Reporting System needs to change asap in order to reduce the errors which affect the patient's report and indirectly patient's health in this way or any other way.

Proposed Solution:

The proposed system is an "Online Diagnostic Lab Reporting System". Its main aim is to bring together various diagnostic working, researches on one single platform that is also online (so that it is accessible for everyone).

- The system allows automate diagnosis system.
- Allows for faster service.
- Allows increased sales and profits for diagnostic labs.
- Easy, user friendly GUI.

Advantages

The advantages of the proposed project are that the online system simply automates diagnosis system. It is faster and reliable, provides better services as

compare to manual system. This system may increase the profit for diagnosis labs. The user friendly and interactive interface makes using this application easy for everyone. There is almost zero percent chance of report swapping or missing which has quite fair chances in manual system. The android version of the proposed application save users to visit website again and again, they can checks all details on finger tips.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

a. Non Functional Requirements

Non-functional requirements are the quality requirements that stipulate how well software does what it has to do. These are Quality attributes of any system; these can be seen at the execution of the system and they can also be the part of the system architecture.

b. Accuracy:

The system will be accurate and reliable based on the design architecture. If there is any problem in the accuracy then the system will provide alternative ways to solve the problem.

c. Usability:

The proposed system will be simple and easy to use by the users. The users will comfort in order to communicate with the system. The user will be provided with an easy interface of the system.

d. Accessibility:

The system will be accessible through internet and there should be no any known problem.

e. Performance:

The system performance will be at its best when performing the functionality of the system.

f. Reliability:

The proposed system will be reliable in all circumstances and if there is any problem that will be affectively handle in the design.

g. Security:

The proposed system will be highly secured; every user will be required registration and username/password to use the system. The system will do the proper authorization and authentication of the users based on their types and their requirements. The proposed system will be designed persistently to avoid any misuse of the application.

IMPLEMENTATION DETAILS

5 IMPLEMENTATION DETAILS

a. Introduction to Html Framework

Hyper Text Markup Language, commonly referred to as HTML, is the standard markup language used to create web pages. Along with CSS, and JavaScript, HTML is a cornerstone technology used to create web pages, as well as to create user interfaces for mobile and web applications. Web browsers can read HTML files and render them into visible or audible web pages. HTML describes the structure of a website semantically along with cues for presentation, making it a markup language, rather than a programming language.

HTML elements form the building blocks of HTML pages. HTML allows images and other objects to be embedded and it can be used to create interactive forms. It provides a means to create structured documents by denoting structuralsemantics for text such as headings, paragraphs, lists, links, quotes and other items. HTML elements are delineated by tags, written using angle brackets. Tags such as <img /> and <input /> introduce content into the page directly. Others such as ... surround and provide information about document text and may include other tags as sub-elements. Browsers do not display the HTML tags, but use them to interpret the content of the page.

HTML can embed scripts written in languages such as JavaScript which affect the behavior of HTML web pages. HTML markup can also refer the browser to Cascading Style Sheets (CSS) to define the look and layout of text and other material.

HyperText Markup Language (HTML) is the standard markup language for creating web pages and web applications. With Cascading Style Sheets (CSS) and JavaScript it forms a triad of cornerstone technologies for the World Wide Web.[1] Web browsers receive HTML documents from a webserver or from local

storage and render them into multimedia web pages. HTML describes the structure of a web page semantically and originally included cues for the appearance of the document. HTML elements are the building blocks of HTML pages. With HTML constructs, images and other objects, such as interactive forms, may be embedded into the rendered page. It provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items. HTML elements are delineated by tags, written using angle brackets. Tags such as <img /> and <input /> introduce content into the page directly. Others such as ... surround and provide information about document text and may include other tags as sub- elements. Browsers do not display the HTML tags, but use them to interpret the content of the page.

HTML can embed programs written in a scripting language such as JavaScript which affect the behavior and content of web pages. Inclusion of CSS defines the look and layout of content. The World Wide Web Consortium (W3C), maintainer of both the HTML and the CSS standards, has encouraged the use of CSS over explicit presentational HTML since 1997.

HTML documents imply a structure of nested HTML elements. These are indicated in the document by HTML tags, enclosed in angle brackets thus:

[61]In the simple, general case, the extent of an element is indicated by a pair of tags: a "start tag" and "end tag" . The text content of the element, if any, is placed between these tags. Tags may also enclose further tag markup between the start and end, including a mixture of tags and text. This indicates further (nested) elements, as children of the parent element.

The start tag may also include attributes within the tag. These indicate other information, such as identifiers for sections within the document, identifiers used to bind style information to the presentation of the document, and for some tags

such as the <img> used to embed images, the reference to the image resource. Some elements, such as the line break <br>, do not permit any embedded content, either text or further tags. These require only a single empty tag (akin to a start tag) and do not use an end tag.

Many tags, particularly the closing end tag for the very commonly used paragraph element <p>, are optional. An HTML browser or other agent can infer the closure for the end of an element from the context and the structural rules defined by the HTML standard. These rules are complex and not widely understood by most HTML coders.

The general form of an HTML element is therefore: <tag attribute1="value1" attribute2="value2">"content"</tag>. Some HTML elements are defined as empty elements and take the form <tag attribute1="value1" attribute2="value2">. Empty elements may enclose no content, for instance, the <br> tag or the inline <img> tag. The name of an HTML element is the name used in the tags. Note that the end tag's name is preceded by a slash character, "/", and that in empty elements the end tag is

neither required nor allowed. If attributes are not mentioned, default values are used in each case.

b. Cascading Style Sheets (CSS)

CSS is a style sheet language used for describing the presentation of a document written in a markup language. Although most often used to set the visual style of web pages and user interfaces written in HTML and XHTML, the language can be applied to any XML document, including plain XML, SVG andXUL, and is applicable to rendering in speech, or on other media. Along with HTML and JavaScript, CSS is a cornerstone technology used by most websites to create visually engaging webpages, user interfaces for web applications, and user interfaces for many mobile applications.

CSS is designed primarily to enable the separation of document content from document presentation, including aspects such as the layout, colors, and fonts. This separation can improve content accessibility, provide more flexibility and control in the specification of presentation characteristics, enable multiple HTML pages to share formatting by specifying the relevant CSS in a separate .css file, and reduce complexity and repetition in the structural content, such as semantically insignificant tables that were widely used to format pages before consistent CSS rendering was available in all major browsers. CSS makes it possible to separate presentation instructions from the HTML content in a separate file or style section of the HTML file. For each matching HTML element, it provides a list of formatting instructions. For example, a CSS rule might specify that "all heading 1 elements should be bold", leaving pure semantic HTML markup that asserts "this text is a level 1 heading" without formatting

code such as a<bold> tag indicating how such text should be displayed.

This separation of formatting and content makes it possible to present the same markup page in different styles for different rendering methods, such as on- screen, in print, by voice (when read out by a speech-based browser orscreen

reader) and on Braille-based, tactile devices. It can also be used to display the web page differently depending on the screen size or device on which it is being viewed. Although the author of a web page typically links to a CSS file within the markup file, readers can specify a different style sheet, such as a CSS file stored on their own computer, to override the one the author has specified. If the author or the reader did not link the document to a style sheet, the default style

of the browser will be applied. Another advantage of CSS is that aesthetic changes to the graphic design of a document (or hundreds of documents) can be applied quickly and easily, by editing a few lines in one file, rather than by a laborious (and thus expensive) process of crawling over every document line by line, changing markup.

The CSS specification describes a priority scheme to determine which style rules apply if more than one rule matches against a particular element. In this so-called cascade, priorities (or weights) are calculated and assigned to rules, so that the results are predictable.

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Readers can also specify a different style sheet, such as a CSS file stored on their own computer, to override the one the author specified.

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The CSS specifications are maintained by the World Wide Web Consortium (W3C). Internet media type (MIME type) text/css is registered for use with CSS by RFC 2318 (March 1998). The W3C operates a free CSS validation service for CSS documents.

c. MYSQL Server

MySQL is an open-source relational database management system (RDBMS);[6] in July 2013, it was the world's second most widely used RDBMS, and the most widely used open-source client–server model RDBMS. It is named after co-founder Michael Widenius's daughter, My. The SQL acronym stands for Structured Query Language. The MySQL development project has made its source code available under the terms of the GNU General Public License, as well as under a variety of proprietary agreements. MySQL was owned and sponsored by a single for- profit firm, the Swedishcompany MySQL AB, now owned by Oracle Corporation. For proprietary use, several paid editions are available, and offer additional functionality.

MySQL (officially pronounced as /maɪˌɛskjuːˈɛl/ "My S-Q-L",[6]) is an open- source relational database management system (RDBMS).[7] Its name is a combination of "My", the name of co-founder Michael Widenius' daughter,[8] and "SQL", the abbreviation for Structured Query Language. The MySQL development project has made its source code available under the terms of the GNU General Public License, as well as under a variety of proprietary agreements. MySQL was owned and sponsored by a single for- profit firm, the Swedish company MySQL AB, now owned by Oracle Corporation.[9] For proprietary use, several paid editions are available, and offer additional functionality.

MySQL is a central component of the LAMP open-source web application software stack (and other "AMP" stacks). LAMP is an acronym for "Linux, Apache, MySQL, Perl/PHP/Python". Applications that use the MySQL database include: TYPO3, MODx, Joomla, WordPress, phpBB, MyBB, and Drupal. MySQL is also used in many high-profile, large-scale websites, including Google[10][11] (though not for searches), Facebook,[12][13][14]Twitter,[15] Flickr,[16] and YouTube.

d. PHP

PHP is a server-side scripting language designed for web development but also used as a general-purpose programming language. Originally created by RasmusLerdorf in 1994, the PHP reference implementation is now produced by The PHP Group. PHP originally stood for Personal Home Page, but it now stands for therecursive backronym PHP: Hypertext Preprocessor.

PHP code may be embedded into HTML code, or it can be used in combination with various web template systems, web content management system and web frameworks. PHP code is usually processed by a PHPinterpreter implemented as a module in the web server or as a Common Gateway Interface (CGI) executable. The web server combines the results of the interpreted and executed PHP code, which may be any type of data, including images, with the generated web page. PHP code may also be executed with a command-line interface(CLI) and can be used to implement standalone graphical applications.

The standard PHP interpreter, powered by the Zend Engine, is free software released under the PHP License. PHP has been widely ported and can be deployed on most web servers on almost every operating system andplatform, free of charge.

The PHP language evolved without a written formal specification or standard until 2014, leaving the canonical PHP interpreter as a de facto standard. Since 2014 work has gone on to create a formal PHP specification.

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Rasmus Lerdorf in 1994,[4] the PHP reference implementation is now produced by The PHP Development Team.[5] PHP originally stood for Personal Home Page,[4] but it now stands for the recursive acronym PHP: Hypertext Preprocessor.[6]

PHP code may be embedded into HTML or HTML5 code, or it can be used in combination with various web template systems, web content management systems and web frameworks. PHP code is usually processed by a PHP interpreter implemented as a module in the web server or as a Common Gateway Interface (CGI) executable. The web server combines the results of the interpreted and executed PHP code, which may be any type of data, including images, with the generated web page. PHP code may also be executed with a command- line interface (CLI) and can be used to implement standalone graphical applications.[7]

The standard PHP interpreter, powered by the Zend Engine, is free software released under the PHP License. PHP has been widely ported and can be deployed on most web servers on almost every operating system and platform, free of charge.[8]

The PHP language evolved without a written formal specification or standard until 2014, leaving the canonical PHP interpreter as a de facto standard. Since 2014 work has gone on to create a formal PHP specification.

e. ANGULAR JAVA SCRIPT

Angular (commonly referred to as "Angular" or "Angular.js") is an open- source web application framework mainly maintained by Google and by a community of individuals and corporations to address many of the challenges encountered in developing single-page applications. It aims to simplify both the development and the testing of such applications by providing a framework for client-side model– view–controller (MVC) and model–view–viewmodel(MVVM) architectures, along with components commonly used in rich Internet applications.

The AngularJS framework works by first reading the HTML page, which has embedded into it additional custom tag attributes. Angular interprets those attributes as directives to bind input or output parts of the page to a model that is represented by standard JavaScript variables. The values of those JavaScript variables can be manually set within the code, or retrieved from static or dynamic JSON resources.

According to JavaScript analytics service Libscore, AngularJS is used on the websites of Wolfram Alpha, NBC,Walgreens, Intel, Sprint, ABC News, and approximately 8,400 other sites out of 1 million tested in July 2015.

AngularJS is the frontend part of the MEAN stack, consisting of MongoDB database, Express.js web application server framework, Angular.js itself, and Node.js runtime environment. AngularJS (commonly referred to as "Angular.js" or "AngularJS 1.X") is a JavaScript-based open-source front-end web application framework mainly maintained by Google and by a community of individuals and corporations to address many of the challenges encountered in developing single-page applications. The JavaScript components complement Apache Cordova, the framework used for developing cross-platform mobile apps. It aims to simplify both the development and the testing of such applications by providing a framework for client-side model–view–controller (MVC) and model–view– viewmodel (MVVM) architectures, along with components commonly used in rich Internet applications. In 2014, the original AngularJS team began working on Angular (Application Platform).

The AngularJS framework works by first reading the HTML page, which has embedded into it additional custom tag attributes. Angular interprets those

attributes as directives to bind input or output parts of the page to a model that is represented by standard JavaScript variables. The values of those JavaScript variables can be manually set within the code, or retrieved from static or dynamic JSON resources.

According to JavaScript analytics service Libscore, AngularJS is used on the websites of Wolfram Alpha, NBC, Walgreens, Intel, Sprint, ABC News, and approximately12,000 other sites out of 1 million tested in October 2016.[3]AngularJS is the 6th most starred project of all time on GitHub.[4] AngularJS is the frontend part of the MEAN stack, consisting of MongoDB database, Express.js web application server framework, Angular.js itself, and Node.js server runtime environment.

AngularJS is built on the belief that declarative programming should be used to create user interfaces and connect software components, while imperative programming is better suited to defining an application's business logic.[5] The framework adapts and extends traditional HTML to present dynamic content through two-way data-binding that allows for the automatic synchronization of models and views. As a result, AngularJS de-emphasizes explicit DOM manipulation with the goal of improving testability and performance.

AngularJS's design goals include: to decouple DOM manipulation from application logic. The difficulty of this is dramatically affected by the way the code is structured. To decouple the client side of an application from the server side. This allows development work to progress in parallel, and allows for reuse of both sides. To provide structure for the journey of building an application: from designing the UI, through writing the business logic, to testing.

Angular implements the MVC pattern to separate presentation, data, and logic components.[6] Using dependency injection, Angular brings traditionally server-side services, such as view-dependent controllers, to client- side web applications. Consequently, much of the burden on the server can be reduced.

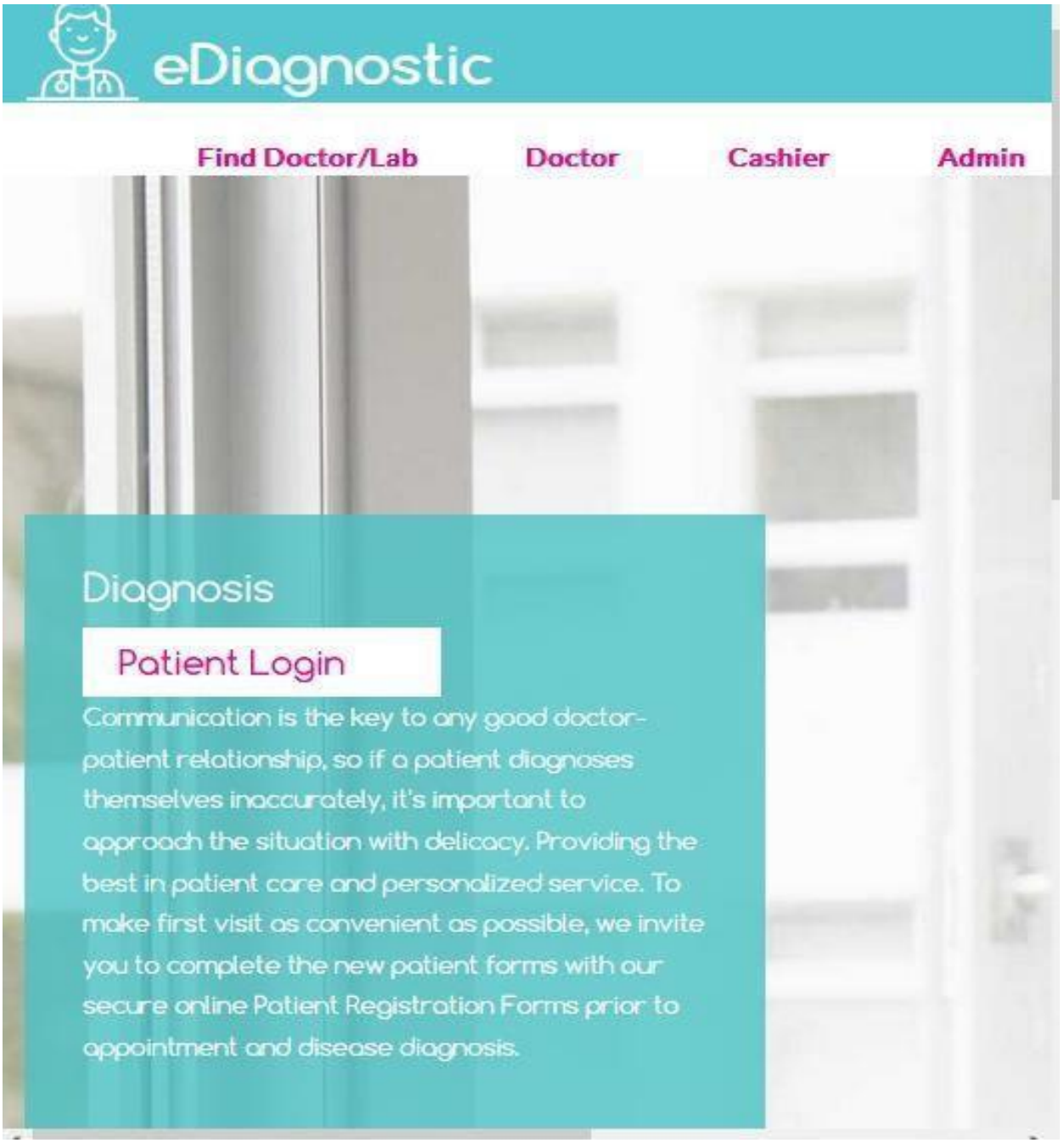
LIST OF REPORT

Input-Output

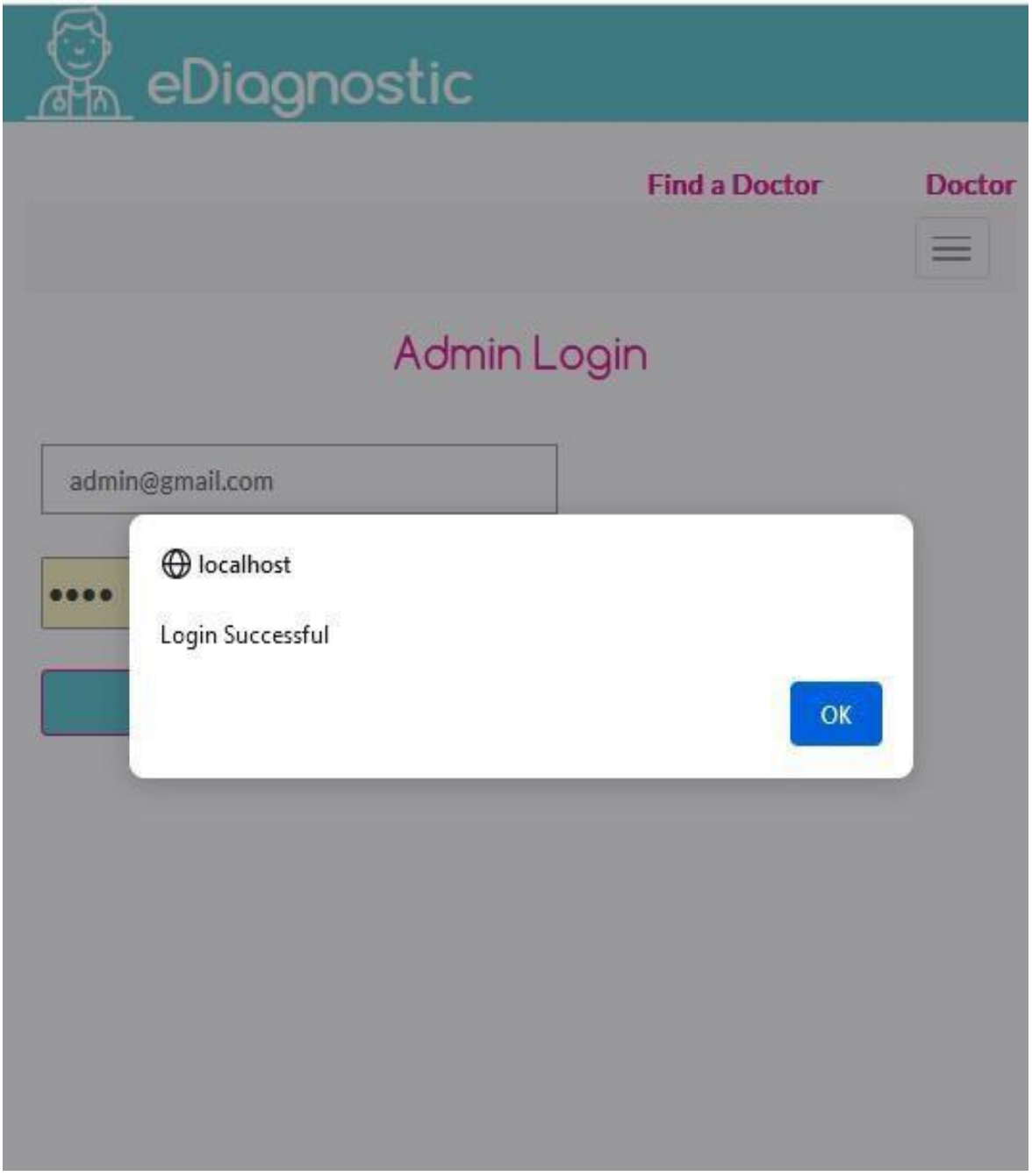
SCREENSHOTS

6 SCREENSHOT

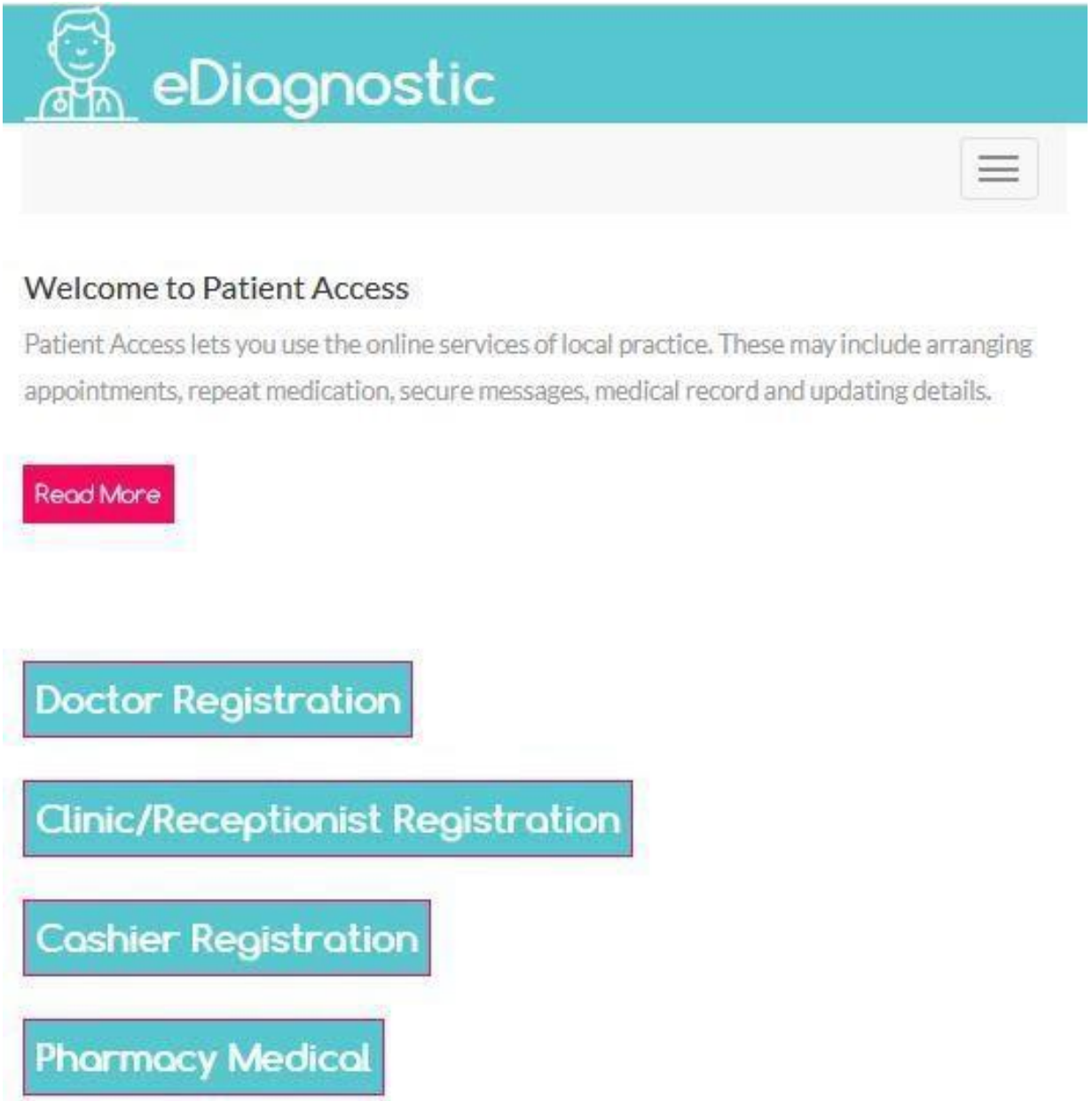
HOME PAGE



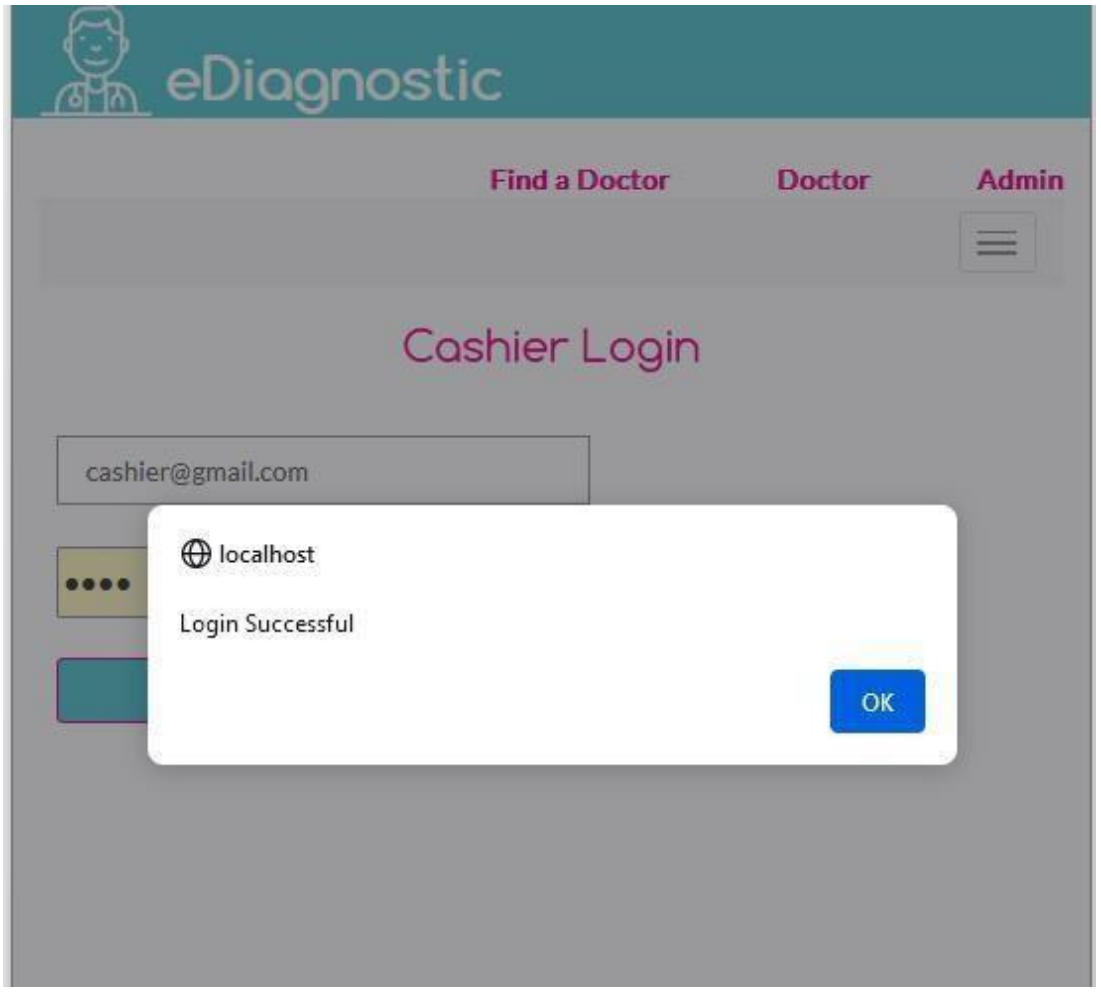
ADMIN LOGIN PAGE



PATIENT ACCESS PAGE



CASHIER LOGIN PAGE



POST PAYMENT PAGE

CONCLUSION

7 CONCLUSION

It serves the needs of Pathology, Bio-chemistry, Microbiology, Hematology, Chemical makeup, Toxicology, X-ray, Sonographer, ECG, CT scan, and MRI confirming labs. It is MS windows Centered, Highly effective application for Diagnostics Center. It is Multi-user,

Multi-tasking system. Effective, Flexible and Confirm Software. Operates on LAN (Network) where you can link up to 150 computer devices at the same time.

In this proposed Diagnostic Lab Reporting System, the program allows improve diagnosis program. Allow faster service. Allows increase in sales for analytic laboratories. Easy to use and simple Graphical User Interface.

REFERENCES

8 REFERENCES

[1] Al-Khawlani, M. A. A. (2009), A Web-Based Integrated Health Care Management System (Master's thesis, The University of Malaya, Kuala Lumpur, Malaysia) Retrieved from: http://repository.um.edu.my/394 on 22/06/2017.

[2] Brailer, D. J. (2005). National Health Information Technology, US Health and Human Services Department, Washington, USA. Economic perspectives on health information technology. The Journal of the National Association for Business Economics.

[3] Cholewka, P. A. (2006). Implementation of a health care information system in Lithuania. International Journal of Economic Development, 8(3), 716-747. Retrieved from: http://www.ijesar.org. on 20/07/2017.

[4] Monu, R. (2010) Design and implementation of a basic laboratory information system for resourcelimited settings (Master's Thesis, Georgia Institute of Technology, Atlanta, Georgia, USA). Retrieved from https://smartech.gatech.edu/bitstream/handle/1853/34792/ on 01/06/2017.

[5] Mustafa, Y. (2004) E-Health Centre: a web-based tool to empower patients to become proactive customers. Health Information and Libraries Journal, 21, 129-133.

[6] Porta-sales, J., Codorniu, N., Gómez-batiste, X., Alburquerque, E., Serrano-Bermúdez, G., SánchezPosadas, D., Pérez-Martin, X., González-Barboteo, J. & Tuca-Rodríguez, A. (2005) Patient Appointment Process, Symptom Control and Prediction of Follow-up Compliance in a Palliative Care Outpatient Clinic. Pain and Symptom Management, 30, 145-153.

Appendices

SOURCE CODE

Index.html

```
<!DOCTYPE html>
<html ng-app="myapp">
<head>
<title>eDiagnostic a Medical Hospital Category Flat Bootstarp Resposive Website Template |
Home ::
</title>
<!-- for-mobile-apps -->
<meta name="viewport" content="width=device-width, initial-scale=1">
<meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
<meta name="keywords" content="eDiagnostic Responsive web template, Bootstrap Web
Templates, Flat Web Templates, Andriod Compatible web template,
Smartphone Compatible web template, free webdesigns for Nokia, Samsung, LG,
SonyErricsson, Motorola web design" />
<script type="application/x-javascript"> addEventListener("load", function()
{ setTimeout(hideURLbar, 0); }, false);
function hideURLbar(){ window.scrollTo(0,1); } </script>
<!-- //for-mobile-apps -->
<link href="css/bootstrap.css" rel="stylesheet" type="text/css" media="all" />
<link href="css/style.css" rel="stylesheet" type="text/css" media="all" />
<!-- js -->
<script src="js/jquery-1.11.1.min.js"></script>
<!-- //js -->
<!-- start-smoth-scrolling -->
<script type="text/javascript" src="js/move-top.js"></script>
<script type="text/javascript" src="js/easing.js"></script>
<script type="text/javascript">
        jQuery(document).ready(function(
        $) {
                $(".scroll").click(function(event){

                        event.preventDefault();
                        $('html,body').animate({scrollTop:$(this.hash).offset().top},1000);

                });

        });
</script>
<!-- start-smoth-scrolling -->
</head>


<body ng-controller="myappCtrl">
<!-- header -->
        <div class="header">
                <div class="container">
                        <div class="logo">
```

```
                        <a href="index.html">eDiagnostic</a>
                    </div>


                    <div class="clearfix"> </div>
                </div>
            </div>
            <div class="header-bottom">
                <div class="container">
                    <div class="header-bottom-right">
                        <ul>

                        <li  class="mail"><a href="find_doctor.html">Find
Doctor/Lab</a></li>

                        <li class="mail" ><a href="doctor_login.html">Doctor</a></li>
                        <li class="mail" ><a href="cashier_login.html">Cashier</a></li>
                            <li class="mail"
><a href="admin_login.html">Admin</a></li>
                        </ul>

                    </div>
                    <div class="clearfix"> </div>

                </div>

            </div>
<!-- //header -->
<!-- banner -->
            <div class="banner">
                <div class="container">
                    <div class="banner-nav">


<!-- Slider-starts-Here -->
                        <script src="js/responsiveslides.min.js"></script>
                         <script>
                           // You can also use "$(window).load(function() {"
                           $(function () {
                             // Slideshow 4
                             $("#slider3").responsiveSlid
                               es({ auto: true,
                               pager:
                               true, nav:
                               false,
                               speed:
                               500,
                               namespace:
                               "callbacks", before:
                               function () {
                                $('.events').append("<li>before event fired.</li>");
```

```
},
after: function () {
 $('.events').append("<li>after event fired.</li>");
}}););
```

```
</script>
<!--//End-slider-script -->

<div id="top" class="callbacks_container wow fadeInUp" data-wow-
delay="0.5s">

<ul class="rslides" id="slider3">

<div class="banner-info">

<h1>Diagnosis</h1><a href="patient_login.html" class="hvr-shutter-out-horizontal"
style="margin-top:0px;margin-right:700px;width:200px">Patient Login</a>

<p>Communication is the key to any good
doctor-patient relationship,

so if a patient diagnoses themselves
inaccurately, it's important to

approach the situation with delicacy.

Providing the best in patient care and
personalized service.

To make  first visit as convenient as
possible,
we invite

you to complete the new patient forms
with
our secure online

Patient Registration Forms prior to
appointment and disease diagnosis.

</p>

</div>

</ul>

</div>
<!--//End-slider-script -->


<div id="top" class="callbacks_container wow fadeInUp"
style="margin- top:0px" data-wow-delay="0.5s">
<ul class="lslides" id="slider3">
```

```
                                                    <div class="banner-info">
            <h1>

                                                    Diagnosis</h1><a
href="clinic_login.html" class="hvr-shutter-out-horizontal" style="margin-top:0px;margin-
right:700px;width:200px">Clinic Login</a>

                                                    <p>Communication is the key to any good
doctor-patient relationship,

                                                    so if a patient diagnoses themselves
inaccurately, it's important to

                                                    approach the situation with delicacy.

                                                    Providing the best in patient care and
personalized service.

                                                    To make  first visit as convenient as possible,
we invite

                                                    you to complete the new patient forms with
our secure online

                                                    Patient Registration Forms prior to
appointment and disease diagnosis.


</p>

            </div>

                        </ul>

                    </div>
                    </div>

                </div>

        </div>
<!-- //banner -->
<!-- banner-bottom -->
```

```
<!-- //banner-bottom -->
<!-- banner-bottom-video -->


<!-- //banner-bottom-video -->
<!-- featured-services -->


<!-- //featured-services -->
<!-- view -->
	<div class="view">
		<div class="container">
			<div class="view-left">
				<p><span>Health Diagnosis Programs</span> Working
Professionals today undergo a lot of stress and strain physically and mentally which is not
good for the body.For all those hard working professionals who slog the whole week, we
have programs designed to give you an all round status of  health.</p>
			</div>
			<div class="view-right">
				<a href="#" class="btn atque btn-primary hvr-shutter-out-
horizontal" role="button">Read More</a></p>
			</div>
			<div class="clearfix"></div>
		</div>
	</div>
<!-- //view -->
```

```
<!-- map -->


<!-- //map -->
<!-- footer -->


<!-- //footer -->
<!-- here stars scrolling icon -->


<!-- //here ends scrolling icon -->
<!-- for bootstrap working -->
                    <script src="js/bootstrap.js"> </script>
                    <script src="js/angular_admin.js"></script>




<!-- //for bootstrap working -->
</body>
</html>
```

User_login.php

```
<?php
/* Following code will match admin login credentials */


//user temp array
$response = array();


// include db connect class
require_once __DIR_. '/db_connect.php';


// connecting to db
```

```
// check for post data
$data = json_decode(file_get_contents("php://input"));
$get_empid = ($data->email);
$get_password = ($data->password);


if(empty($get_empid) || empty($get_password))
{

        $response["success"] = 2;  echo json_encode($response);

}
els
e
```

```
{$result = mysqli_query($conn,"SELECT * FROM login WHERE email = '$get_empid' AND
password = '$get_password'");


                if (mysqli_num_rows($result))
                {
                        $Allresponse = mysqli_fetch_array($result);
                        // temp user array
                        $response = array();
                        $response = $Allresponse;

                        $response["success"] = 1;
                        echo
                        json_encode($response);
                } else
                {

                        // success
                        $response["success"] = 0;

                        // echoing JSON response
                        echo
                        json_encode($response);
                }

}
?>
```
[7]

CANDIDATE DECLARATION

Project Title **INTEGRATED AI POWERED PARKING MANAGEMENT SYSTEM**

I , **S SRIRAM**

(Enrolment number: **EA2332251010362**), declare that this project is my own work, based on my personal study and/or research and that I have acknowledged all material and sources used in its preparation, whether they be books, articles, reports, lecture notes, and any other kind of document, electronic or personal communication. I also certify that this project has not previously been submitted for assessment in any academic institution, and that I have not copied in part or whole or otherwise plagiarized the work of other persons. I confirm that I have identified and declared all possible conflicts that I may have.

Signed and submitted

Date: **20/05/2025**